App User Behavior Segmentation

Unsupervised learning with K-Means

Prepared for Guru | Based on the supplied project brief and dataset

This project groups users by similar app activity, identifies the users in each group, and suggests actions for marketing, product, and retention teams. It follows the 10 requested steps using simple Python analysis.

Users	Original columns	Model inputs	Project groups
50,000	25	7 numeric features	4 clusters

Main result

Every user received a cluster assignment. The strongest differences are session duration, pages viewed, and login recency. Average engagement and supplied churn-risk scores are almost the same across groups.

Important finding: the sampled silhouette score is **0.104**, which indicates substantial overlap. Four groups are used for the project requirement; the data does not establish four clearly separated engagement types.

1. Data Collection

The analysis uses **app_user_behavior_dataset.csv**, supplied with the project. Each row represents one user. It contains demographics, device details, app activity, subscription information, and supplied engagement and churn-risk scores. No additional dataset was downloaded.

What this report covers

Data understanding and cleaning; exploration; feature selection and scaling; K-Means and the elbow method; cluster assignment; PCA visualization; user profiles; and business-action mapping.

Scope: descriptive analysis of this dataset snapshot. This project does not establish actual churn, customer financial value, campaign uplift, or production readiness.

Understand and clean the data

2. Data Understanding

Column group	Examples	Purpose
Identity and demographics	user_id, age, gender, country	Identify and describe users
Device and account	device_type, app_version, account_age_days	Describe technical and account context
Usage behavior	sessions_per_week, avg_session_duration_min, daily_active_minutes	Measure frequency and time spent
Interaction and recency	feature_clicks_per_session, pages_viewed_per_session, days_since_last_login	Measure interaction depth and last activity
Scores and commercial context	engagement_score, churn_risk_score, subscription_type, rating_given	Profile groups after modeling

Column meanings are inferred from names. The formulas for the supplied scores are unavailable. A churn-risk score is not an observed churn event. A software version is treated as a category, not a numerical quantity.

3. Data Cleaning

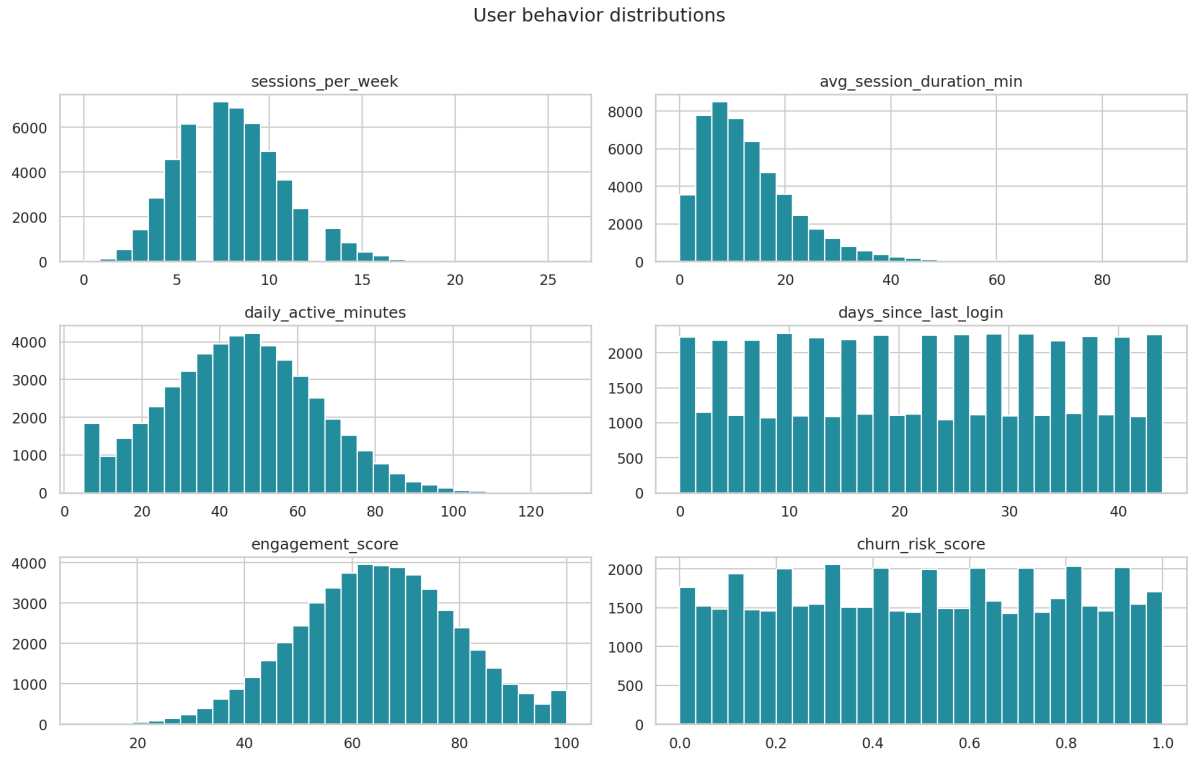
Check	Result / treatment
Missing ratings	5,022 of 50,000 users (10.04%). Filled with median rating 4; preserved rating_was_missing.
Other missing values	None in the supplied dataset.
Duplicates	0 exact duplicate rows and 0 duplicate user IDs.
Invalid numeric checks	No negative or non-finite values, or out-of-range ratings / supplied scores, found by the implemented checks.
Categories and outliers	Trimmed category spaces; inspected unusual activity values using IQR. Retained plausible heavy users.

Why median? It is the middle value and is less sensitive to unusually large values. Rating is excluded from clustering, so filled ratings do not determine user groups.

Cleaning checks do not prove every field is correct. The time windows of activity metrics should be confirmed with the data owner before interpreting possible contradictions between session frequency and last login.

Explore user behavior

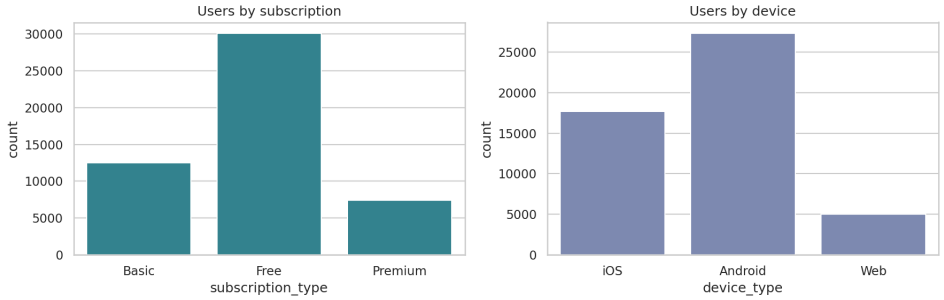
Exploratory Data Analysis (EDA) means looking at distributions and summaries before fitting a model. A histogram counts how many users fall into each value range.



What the charts show

Sessions: users average about 8 sessions per week. **Duration:** average session length is 13.15 minutes, with a long right tail. **Daily activity:** the average is 45.16 minutes.

Login recency: values span 0 to 44 days, with an average of 22.04 days. **Engagement:** the average supplied score is 64.94. **Churn risk:** the supplied score averages 0.501; its calibration is unknown.



Subscription and device categories help describe the audience but are not used in this first behavior-focused clustering model. Differences in counts do not establish differences in engagement.

Choose features and build the model

4. Feature Selection

Selected feature	What it captures
sessions_per_week	How often users start sessions
avg_session_duration_min	How long sessions last
daily_active_minutes	Daily activity volume
feature_clicks_per_session	Feature interaction depth
pages_viewed_per_session	Browsing depth
days_since_last_login	How recently users were active
engagement_score	The supplied engagement measure

User ID is excluded because it is an identifier. Demographics, categories, ratings, and extra counts are excluded to keep this baseline simple. Churn risk is excluded from training and summarized afterward. The engagement-score formula is unknown; future work should compare results with and without it.

5. Data Scaling

StandardScaler subtracts each feature mean and divides by its standard deviation. This makes units comparable so a measurement with larger numbers does not dominate K-Means distances. It does not make all distributions normal or guarantee equal business importance.

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df[features])
```

6. Clustering Model Selection

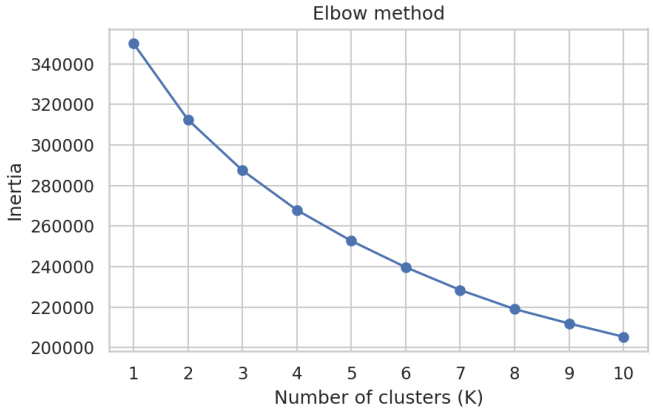
K-Means finds K cluster centers and assigns each user to the nearest center. We use **random_state=42** for repeatability and **n_init=10** to try ten starting arrangements.

All users are used because this is descriptive segmentation of the current snapshot. There is no target label and no classification accuracy. For future-user evaluation, preprocessing must be fitted only on the training data.

Compare the number of clusters

7. Optimal Cluster Identification

The elbow method plots the number of groups (K) against **inertia**: the total squared distance from users to their group centers. Look for a bend where improvement starts slowing. Lower inertia by itself is not enough, because inertia always falls as K increases.





K	Inertia	Sampled silhouette
1	350,000	Not defined
2	312,411	0.1127
3	287,438	0.1114
4	267,870	0.1035
5	252,651	0.1026
10	205,356	0.1028

K=1 through 10 were evaluated. Silhouette was calculated on a reproducible random sample of 3,000 users to reduce computation. Higher is better; a value near zero indicates overlapping groups.

Decision: K=2 has the highest sampled silhouette among the tested values, but it is still low. There is no compelling elbow. **K=4 is retained as the requested project configuration, not a proven statistical optimum.**

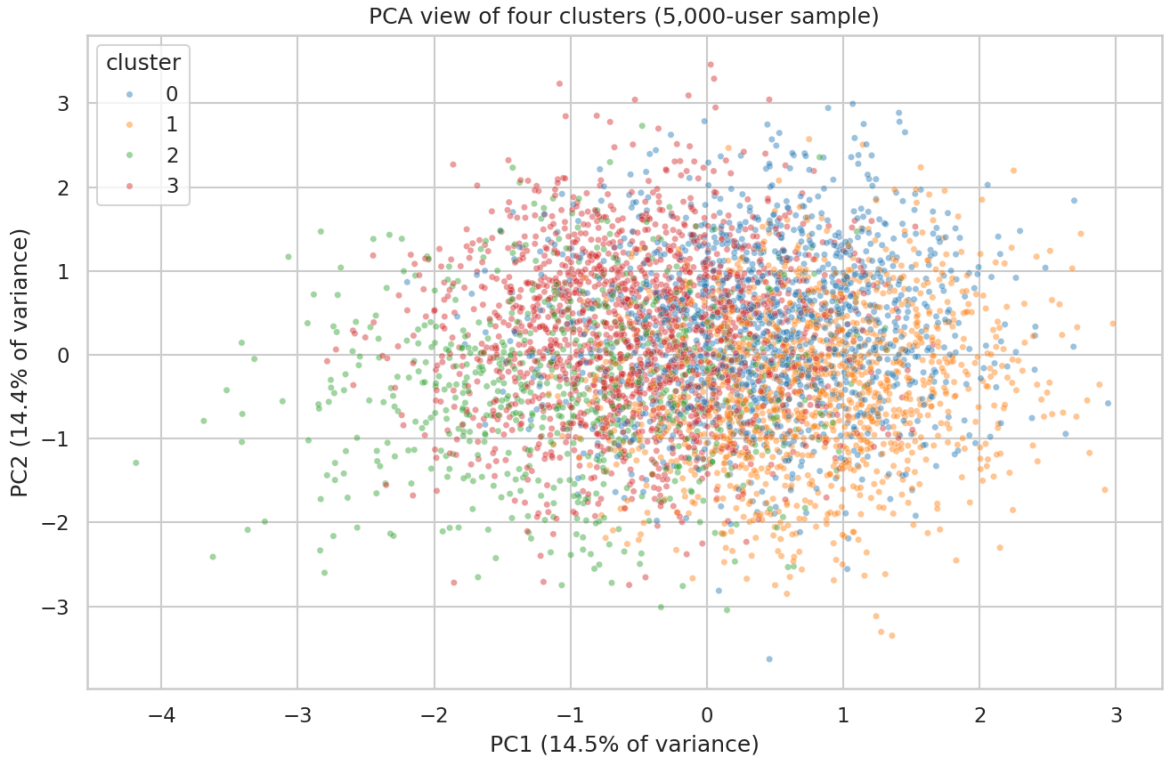
This is a useful result: a clustering method can partition data even when strong natural customer segments are absent.

Assign clusters and inspect separation

8. Cluster Assignment

```
model = KMeans(n_clusters=4, random_state=42, n_init=10)
df["cluster"] = model.fit_predict(X_scaled)
```

Each user receives one cluster number from 0 to 3. These numbers are identifiers, not engagement rankings. PCA is applied afterward to visualize the seven standardized features in two dimensions.



Measure	Result	Meaning
Sampled silhouette	0.1035	Substantial overlap between the four groups
Davies-Bouldin	2.2004	Lower is better; a descriptive separation measure
Two-component PCA variance	28.86%	The 2D view omits much of the original information

How to read the plot: each dot is one sampled user. Colors show assigned clusters. Overlap is visible, but the seven-feature silhouette is the more direct numerical evaluation of the fitted partition. A colored PCA plot alone does not validate cluster quality.

The plot shows a 5,000-user sample. K-Means was fitted on all 50,000 users, using the seven standardized features, not the PCA coordinates.

Identify and profile each group

9. Cluster-Level User Identification and Profiling

ID	Provisional label	Users	Share	Observed pattern
0	Occasional	13,332	26.66%	Less recent logins; more pages per session
1	Low Engagement / At-Risk	13,404	26.81%	Less recent logins; fewer pages per session
2	Moderate Engagement	7,101	14.20%	Longer sessions
3	High Engagement	16,163	32.33%	More recent logins

Naming rule: after fitting, rank the cluster means of an activity index: the average of standardized sessions, daily minutes, engagement, and reversed days since login. The highest-to-lowest ranks receive High, Moderate, Occasional, and Low / At-Risk labels.

These are relative project labels. “Occasional” is not proof of irregular visits, and “At-Risk” is not a churn prediction. Clusters 0 and 1 have almost identical activity-index averages, making their name distinction especially weak.

ID	Session minutes	Pages / session	Days since login	Engagement	Churn score
0	10.42	19.49	30.60	64.82	0.500
1	10.57	7.51	30.69	64.55	0.498
2	29.67	13.76	21.00	65.15	0.498
3	10.29	13.32	8.26	65.27	0.505

Interpretation: cluster 2 has much longer sessions; cluster 3 has more recent logins. Clusters 0 and 1 differ mainly in page views. Average engagement and churn scores remain very similar.

Customer-level outputs

The all-user file contains user ID, original attributes, cluster, provisional label, observed pattern, review flags, and recommended action. Separate files identify the exact users in each cluster.

Validation: the four cluster lists contain 50,000 unique user IDs in total, with no duplicate membership or omitted user. Cluster counts sum to the full dataset size.

Map insights to practical actions

10. Business Insight Generation and Customer Action Mapping

Provisional segment	Suggested experiment
High Engagement	Review eligible users for loyalty or premium offers.
Moderate Engagement	Test relevant content and feature tips.
Occasional	Test a gentle re-engagement reminder; verify visit patterns.
Low Engagement / At-Risk	Investigate barriers and test onboarding support.

Individual review rules

Retention review: supplied churn score >= 0.70, OR engagement < 50 and last login >= 30 days ago. **17,229 users** match this illustrative rule.

Loyalty candidate: engagement >= 80 and sessions per week >= 10. **2,210 users** match; after excluding retention-review overlaps, **1,542 users** remain in the loyalty review list. Retention review takes priority.

These thresholds are manually chosen business rules, not learned or validated risk cutoffs. Without revenue or profit data, engaged users cannot be called proven high-value customers. Actions should be tested before broad use.

Conclusion

The project completes the requested workflow and provides usable customer lists. However, it does **not** confirm four distinct engagement or churn-risk types. Present the observed behavior differences and the weak separation together.

Next improvements

Obtain definitions for the supplied scores and activity time windows. Collect dated session histories and actual churn outcomes. Compare features with and without the engagement score, check cluster stability over time, and measure campaign impact with randomized tests.

Sources and reproducibility

Sources: user-supplied **App User Behavior Segmentation.pdf** and **app_user_behavior_dataset.csv**. Results: the executed **App_User_Segmentation_Beginner.ipynb**, cluster_summary.csv, cluster_evaluation.csv, and user-level exports produced for this project. No external data or benchmarks are used. Values are rounded for presentation.